RESEARCH ARTICLE

Effect of Seasonality and Associated Environmental Drivers on Macroinvertebrate Assemblages in Neotropical Coastal Headwater Streams

Wilson Lafuente[1,2] | José Luis Moreno[3] | Antonio J. Carpio[4]

[1]Departamento de Zoología, Universidad de Córdoba, Córdoba, Spain | [2]Centro de Agua y Desarrollo Sustentable, Escuela Superior Politécnica del Litoral, ESPOL, Guayaquil, Ecuador | [3]Departamento de Producción Vegetal y Tecnología Agraria, Universidad de Castilla-La Mancha, Albacete, Spain | [4]Departamento de Botánica, Ecología y Fisiología Vegetal, Grupo de Investigación en Educación y Gestión de la Biodiversidad (GESBIO), Universidad de Córdoba, Córdoba, Spain

Correspondence: José Luis Moreno (joseluis.moreno@uclm.es)

Received: 31 July 2024 | **Revised:** 25 January 2025 | **Accepted:** 28 January 2025

Funding: This work received financial support from the Agencia Andaluza de Cooperación Internacional para el Desarrollo AACID (projects EMEP 2022UI007_2022). AJC was supported by a "Juan de la Cierva" contract (IJC2020-042629-I) funded by MCIN/AEI/10.13039/501100011033 and by the European Union Next Generation. WL was supported by a mobility grant (PMA1-2024-027-14) funded by the Asociación Universitaria Iberoamericana de Postgrado (AUIP) and the Consejería de Transformación Económica, Industria, Conocimiento y Universidades de la Junta de Andalucía.

Keywords: abiotic factors | benthic invertebrates | diversity | functional feeding groups | neotropical streams | richness

ABSTRACT

Comprehensive research into the environmental factors that affect the spatial and temporal variability of benthic macroinvertebrate communities in the headwater coastal streams of Ecuador is limited. Therefore, a study was carried out in four headwater coastal streams to address this gap. This study lasted for 2 years and included two surveys in summer and two in winter. The spatial and temporal variation of community metrics (abundance, richness, and diversity), functional feeding groups, and environmental factors were analyzed. A total of 47 genera from 33 families and 12 orders were collected, 29 of which were new records for the studied region. Seventy-nine percent of the total number of macroinvertebrates collected belonged to three orders of insects: hemipterans, ephemeropterans and coleopterans. The most abundant non-insect was the amphipod *Hyalella*. In seasonal terms, the populations of odonates, hemipterans, and coleopterans decreased in winter, whereas those of amphipods, trichopterans, ephemeropterans, and dipterans increased. The dominant functional feeding group consisted of predators, with the heteropterans *Rhagovelia* and *Husseyella* being the most abundant genera across all streams. Populations of predators and scrapers decreased in winter, whereas those of shredders and collectors increased. The results showed significant differences among the macroinvertebrate metrics obtained for each stream and season. The abiotic PCA ordination and biotic NMDS ordination of macroinvertebrate assemblages were consistent. The BEST routine inferred that the environmental variables that best explained assemblage patterns were phosphate concentration and stream width, although dissolved oxygen, water temperature, and current velocity were also important explanatory variables. In conclusion, seasons had a major influence on the macroinvertebrate assemblages of neotropical coastal headwater streams. During summer (the dry season), environmental stability supported higher richness and diversity, whereas in winter (the rainy season), hydrologic perturbation disrupted and altered the structure and composition of the macroinvertebrate community.

1 | Introduction

The macroinvertebrate communities in streams play a crucial role in aquatic ecosystems, acting as indicators of environmental health and contributing to nutrient cycling (Wallace and Webster 1996; Crespo-Pérez et al. 2020; Sumudumali and Jayawardana 2021; Tampo et al. 2021; Mariani-Ríos, Maldonado-Benítez, and Ramírez 2022; Hu et al. 2022). These communities are influenced by various human pressures, such as land use changes, flow regulation, reduction of riparian vegetation, and the introduction of invasive species, as well as natural drivers, including the hydrological regime, water chemistry, food resources, and habitat heterogeneity (Chi et al. 2017; Reid et al. 2019; Ríos-Touma and Ramírez 2019; Torremorell et al. 2021; Lim and Do 2023). This is particularly relevant for tropical streams, which experience significant seasonal variations in rainfall, influencing the relative importance of abiotic and biotic factors in structuring the fauna (Jacobsen and Encalada 1998; Couceiro, Dias-Silva, and Hamada 2021; Cortés-Guzmán, Alcocer, and Planas 2022; Da Silva Giehl et al. 2022; Barboza, Tagliacollo, and Jacobucci 2024). Therefore, identifying the main environmental factors that affect the diversity and composition of macroinvertebrate assemblages is essential, with the specific aim of distinguishing between natural and human-induced changes (Pearson et al. 2017).

Studies in Ecuadorian lotic ecosystems have primarily focused on the composition and structure of macroinvertebrate assemblages and the influence of environmental variables on these biotic communities (Sinche et al. 2023; Cabrera-García et al. 2023; Hampel et al. 2023; Vargas-Tierras et al. 2023). Most of these studies have been carried out in highland streams, addressing topics such as the structure and diversity of invertebrate assemblages in streams as related to altitude (Jacobsen, Schultz, and Encalada 1997; Sites, Willig, and Linit 2003; Jacobsen 2004; Kuhn et al. 2011; Ríos-Touma, Encalada, and Prat Fornells 2011; Crespo-Pérez et al. 2016; Villamarín, Rieradevall, and Prat 2020; Villamarín et al. 2021; Ríos-Touma et al. 2022). However, there are fewer studies focusing on macroinvertebrates in streams from the eastern and western regions of Ecuador (e.g., Martínez-Sanz et al. 2014; Lessmann et al. 2016; Urdanigo et al. 2019; Galarza et al. 2021). Research in the western region has mainly evaluated the impact of human pressures on aquatic ecosystems through the use of biotic indices based on benthic macroinvertebrate communities (Dominguez-Granda, Lock, and Goethals 2011; Damanik-Ambarita et al. 2016; Nguyen et al. 2017; Molinero et al. 2019).

However, macroinvertebrate communities inhabiting the headwater coastal streams of Ecuador are rarely studied. Therefore, the objective of this study was to contribute initial insights into the natural factors influencing the structure and distribution of macroinvertebrate communities in these streams. The main objectives of this research were (1) to describe the composition and structure of the macroinvertebrate community; (2) to analyze the spatial (among streams) and temporal (across seasons) variability; and (3) to determine the environmental drivers influencing the variability of macroinvertebrate assemblages.

2 | Materials and Methods

2.1 | Study Area

This study was conducted in four small headwater streams (Buenos Aires, Santa Rosa, San Carlos, and Pozo Hondo) located in the central-southern region of the Ecuadorian coast (Figure 1). According to Strahler (1957), they were first- and second-order streams with lengths ranging from 5 to 10 km and flowing in a mountainous area (426 to 647 m.a.s.l.). They were characterized by dense riparian forest and a high canopy. The predominant land use in the study watersheds was woodland, with some surface dedicated to the cultivation of maize crops. No reports of domestic or industrial wastewater discharges have been documented for the streams studied (Lafuente et al. 2023). The rainy season in the region is from December to May (winter), while the dry season spans from June to November (summer) (Escribano-Ávila 2016; Nguyen et al. 2017). The average monthly air temperature is 25.7°C, and the mean annual precipitation was 568.3 mm for the period 2010–2018 (data obtained from the Ecuadorian National Institute of Meteorology and Hydrology). The climate in the region ranges from tropical semi-humid megathermal (Buenos Aires stream) and tropical semi-arid megathermal (Santa Rosa stream), to tropical dry megathermal (San Carlos and Pozo Hondo streams) (ODEPLAN, 2004).

2.2 | Sites and Sampling Period

The survey was planned in a seasonal manner, with two sampling campaigns in winter and two in summer: September – October 2020, April –May 2021, September –October 2021, and March –April 2022. Four sites were established for each stream, with an approximate distance of 100 m between each site, amounting to a total of 16 sites that were sampled on four dates (Figure 1). Macroinvertebrates samples were collected on each site and date, and the physicochemical and hydromorphological variables were measured.

2.3 | Stream Water Quality and Hydromorphological Variables

A basic set of physicochemical parameters of stream water quality was measured on each site and date ($n = 64$ records). This set included parameters such as temperature (°C), pH, dissolved oxygen (DO) (mg/L), electrical conductivity (EC) (μS/cm), and turbidity (NTU). We also analyzed nitrite-nitrogen (USEPA diazotization method, detection limit of 0.002–0.300 mg/L $N-NO_2^-$), nitrate-nitrogen (cadmium reduction method, detection limit of 0.01–0.50 mg/L $N-NO_3^-$), orthophosphate (USEPA ascorbic acid method, detection limit of 0.02–2.50 mg/L PO_4^{3-}), and nitrogen-ammonia (salicylate method, detection limit of 0.01–0.50 mg/L $N-NH_3$). The hydromorphological variables included the average current velocity, width, and depth.

2.4 | Benthic Macroinvertebrate Communities

A benthic macroinvertebrate sample was collected on each site and date immediately after measuring the water quality

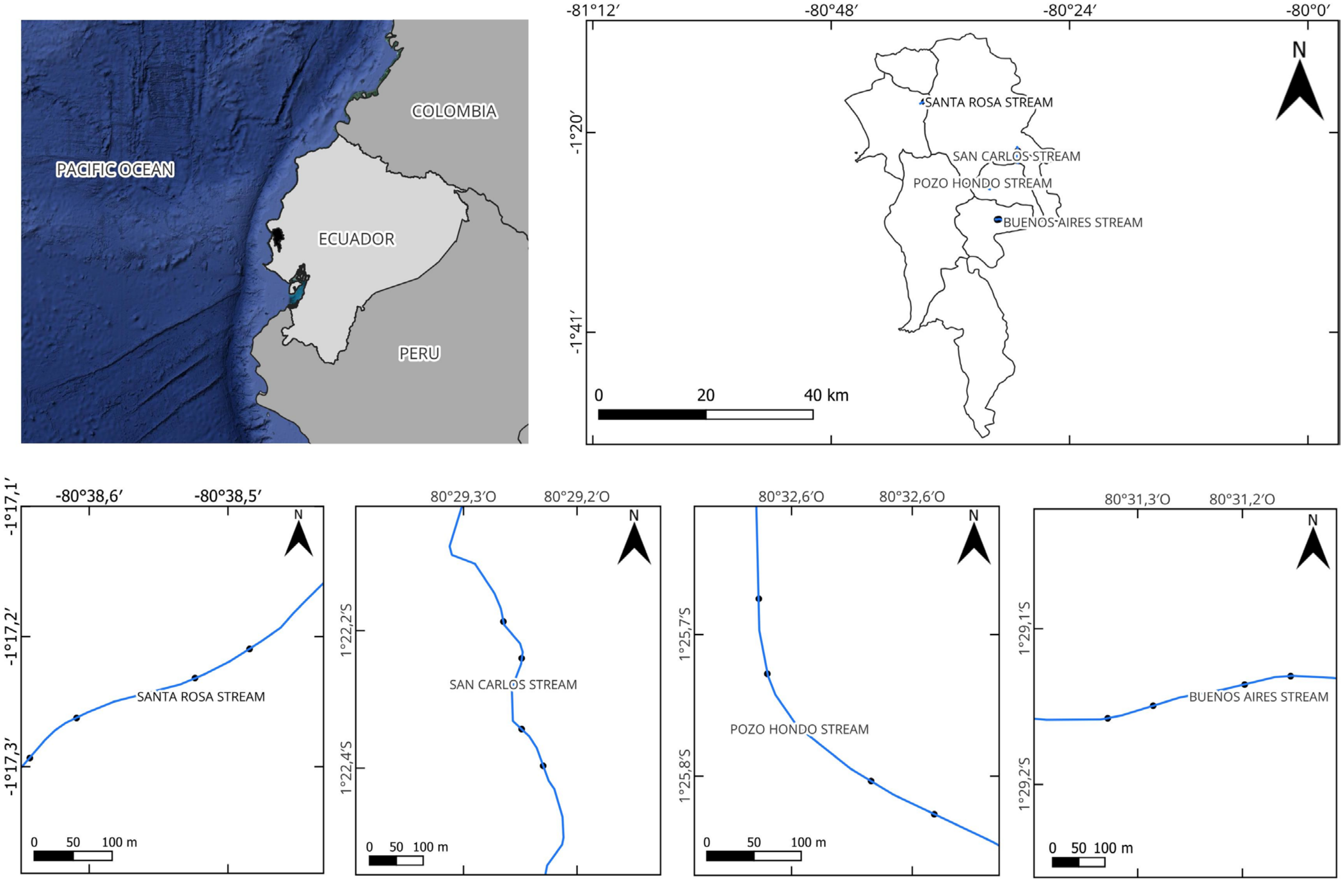

FIGURE 1 | Study area showing the location of the four Ecuadorian coastal headwater streams and sample sites. [Color figure can be viewed at wileyonlinelibrary.com]

variables, amounting to a total of 64 samples during the study period. A standard hand net, consisting of a conical net with a pore size of 500 μm attached to a metal frame, was used for collection. Active sampling was conducted for 3 min and included all the different microhabitats of the stream (Gabriels et al. 2010), ensuring that the accumulated sampling area was equivalent for each sample. The samples collected were stored in plastic bags and preserved with 70% ethanol for later transfer to the laboratory. In the laboratory, all the macroinvertebrates were sorted manually, subsequently counted and identified to the genus level with the help of a Nikon SMZ745T stereomicroscope and taxonomic identification keys (Roldán 1988; Domínguez and Fernández 2009; Hamada, Thorp, and Rogers 2018). Each of the taxa identified was assigned to a specific functional feeding group (FFG) (according to Ramírez and Gutiérrez-Fonseca 2014; Damanik-Ambarita et al. 2016; Van Echelpoel et al. 2018; and Cabrera et al. 2021).

2.5 | Statistical Analysis

A Principal Component Analysis (PCA) was performed on the sample-environmental matrix in order to explore environmental gradients across streams and seasons, reducing the multidimensional representation of samples (12 environmental variables) to two-dimensional ordinations of samples. Environmental variables were standardized prior to running the PCA, using the "normalize" function in PRIMER v7 software to ensure comparability across variables with different units and scales. The resulting components of the PCA are linear combinations of the environmental variables making up the components. The PCA was performed using the PRIMER v7 software (Clarke and Gorley 2006). Three models were carried out: one GLM (General Linear Model), in which the response variable was diversity (Model 1), and two GzLM (Generalized Linear Model) in which total abundance (Model 2), and taxon richness (Model 3) were the dependent variables. A normal distribution and an identity-link function were used in Model 1, while a Poisson distribution and log–link function were used in Models 2 and 3. Streams (four levels) and the climatic season (two levels) were included as fixed factors in these models. The best model was selected by employing the forward-stepwise procedure of selecting the model with the lowest corrected Akaike information criterion (AICc) value. These analyses were carried out using the InfoStat software, updated in 2020 (Di Rienzo et al. 2011). Prior to building the models, an analysis of the collinear environmental variables was carried out using the IBM SPSS Statistics 29 software to remove variables with Variation Inflation Factors (VIF) > 3 (Zuur, Ieno, and Smith 2007; Zuur et al. 2009). Furthermore, before statistical analysis, macroinvertebrate taxa with fewer than five individuals (0.46% of the total sample) were collected throughout the study were eliminated (Jackson, Battle, and Sweeney 2010).

A Permutational Multivariate Analysis of Variance (PERMANOVA) was performed to evaluate the spatial and

TABLE 1 | Mean values and standard deviation (sd) of physicochemical and hydromorphological variables for the streams studied across both seasons.

		Buenos Aires	Pozo Hondo	San Carlos	Santa Rosa	Summer	Winter
Nitrate (mg/L)	Mean	0.08	0.17	0.03	0.10	0.11	0.10
	sd	0.08	0.11	0.03	0.09	0.12	0.10
Ammonia (mg/L)	Mean	0.09	0.11	0.10	0.10	0.10	0.10
	sd	0.02	0.03	0.04	0.02	0.03	0.03
Nitrite (mg/L)	Mean	0.006	0.009	0.004	0.006	0.006	0.006
	sd	0.003	0.004	0.002	0.003	0.003	0.004
Phosphate (mg/L)	Mean	0.63	1.00	1.85	0.61	1.13	0.84
	sd	0.12	0.70	1.01	0.50	1.05	0.55
pH	Mean	8.09	8.52	8.59	8.48	8.48	8.24
	sd	0.50	0.40	0.49	0.58	0.58	0.51
Dissolved oxygen (mg/L)	Mean	6.10	3.76	4.35	4.86	3.48	6.10
	sd	0.92	1.84	2.36	1.80	1.51	1.12
Conductivity (μS/cm)	Mean	193.75	1156.97	402.34	414.13	420.78	489.55
	sd	83.78	268.39	87.05	198.94	293.68	413.41
Turbidity (NTU)	Mean	19.84	22.24	10.15	26.08	14.09	23.26
	sd	8.44	29.18	3.96	32.32	5.71	30.23
Water temperature (°C)	Mean	21.08	22.15	22.24	20.84	20.79	21.86
	sd	1.03	0.83	0.68	0.82	1.15	0.69
Current velocity (cm/s)	Mean	33.56	56.82	59.35	93.02	0.36	21.30
	sd	11.27	19.93	23.07	21.15	0.15	25.40
Stream depth (cm)	Mean	3.00	3.81	5.73	4.86	3.50	4.25
	sd	1.36	1.61	2.95	4.06	1.92	3.11
Stream width (cm)	Mean	36.23	56.03	152.98	70.47	48.53	88.78
	sd	23.41	30.33	163.23	26.65	30.46	107.22

temporal dissimilarity among the macroinvertebrate assemblages (abundance data matrix). Two fixed factors were considered: stream (four levels: San Carlos, Santa Rosa, Pozo Hondo, and Buenos Aires), and season (two levels: summer and winter). The PERMANOVA was performed using the PRIMER v6 software (Clarke and Gorley 2006), including the PERMANOVA+ add-on package (Anderson 2008). In order to increase the power and precision of the analysis, all tests were based on 9999 permutations (Anderson 2008). The permutation approach has the advantage that it is "distribution free" and is not limited by the typical assumptions of parametric statistics (Walters and Coen 2006). The similarities between macroinvertebrate samples were analyzed by performing a Non-metric Multidimensional Scaling (NMDS) ordination on the log-transformed abundance matrix, using the Bray–Curtis similarity index as a measure of resemblance among samples (resemblance matrix). The NMDS was performed using the PRIMER v7 software (Clarke and Gorley 2006). Environmental and taxon vectors were overlaid on the biotic NMDS to explore abiotic gradients and the most important genera contributing to the ordination of the samples. Vectors represented the Pearson correlation coefficients between variables and the NMDS scores of the samples. Finally, a BEST routine was performed using PRIMER v7 software to discover the best match between the multivariate among-sample patterns of an assemblage and those of the environmental variables associated with the samples, identifying the *best* explanatory variables that optimize that match (Clarke and Gorley 2006).

3 | Results

3.1 | Physicochemical and Hydromorphological Characteristics of Streams

The four headwater streams were small and shallow, ranging from 3 to 5.73 cm in depth and 36.23 to 152.98 cm in width. They were warm (20.84°C–22.24°C), low oxygenated (3.76–6.1 mg/L), alkaline (8.09–8.59 pH), low-nitrogen (0.03–0.17 mg/L nitrogen-nitrate, 0.09–0.11 mg/L ammonia, 0.004–0.009 mg/L nitrogen-nitrite) but high-phosphate (0.61–1.85 mg/L) and mineralized (193.75–1156.97 μS/cm) streams.

The streams differed in the values of their physicochemical and hydromorphological variables (Table 1). Pozo Hondo was the most mineralized stream and had the highest nitrogen concentration. San Carlos was the largest stream (the only second-order stream), with the highest channel width and depth, as well as the highest phosphate values. Buenos Aires was the smallest stream, the least mineralized, the most oxygenated, and had the slowest flow. Santa Rosa, the largest of the three first-order streams, had the fastest flow and the most turbid waters. Seasonal patterns were marked by precipitation cycles, with an increase in discharge, turbidity, and current velocity in winter, contrasting with low flow in summer (Table 1). However, there was a general depletion of oxygen in summer, along with a slight increase in nutrient concentrations (nitrate and phosphate). The water temperature remained virtually constant throughout the year, although there was a slight increase in winter (Table 1).

The environmental gradients of the sites were explored by means of a PCA of the sample-environmental data matrix. The first three components (axes) accounted for 49.3% of the total variation (Table 2): PCA1 = 19.3%, PCA2 = 15.8%, and PCA3 = 14.2%. Due to the similar eigenvalues and variation explained by PCA2 and PCA3, and with the aim to increase the cumulative variance value, two ordination plots were generated: PCA1–PCA2 (Figure 2a) and PCA1–PCA3 (Figure 2b). PCA1 represented a gradient of conductivity (eigenvector = 0.54, Table 2), with the summer and winter samples of Pozo Hondo located on the positive side of the ordination, and the winter samples of Buenos Aires and San Carlos on the negative side. PCA1 was also correlated with a higher concentration of nitrogen (ammonia and nitrate), pH, and water temperature (Table 2). The second axis, PCA2, represented a gradient of stream size (width eigenvector = 0.53, Table 2), with the winter samples of the second-order San Carlos stream positioned on the positive side, and the other first-order streams positioned on the negative side (Figure 2a). The third axis, PCA3, represented a seasonality gradient, with the winter samples positioned on the positive side and the summer samples on the negative side (Figure 2b). Winter (the rainy season) was identified as having conditions of higher turbidity, current velocity and DO, whereas the highest concentration of phosphates was recorded in summer (the dry season), especially for the San Carlos and Pozo Hondo streams (see also Table 1).

TABLE 2 | Results of the Principal Component Analysis (PCA) performed on the environmental data, showing the percentage of variation explained by the first three axes (eigenvalues) and the coefficients for environmental variables in the linear combination making up the PCs (eigenvectors).

Component	Eigen values	%variation	Cumulative %variation
PC1	2.32	19.3	19.3
PC2	1.9	15.8	35.1
PC3	1.7	14.2	49.3

Eigenvectors (variable coefficients)

Variable	PC1	PC2	PC3
NO3	0.320	−0.302	0.201
NH3	0.381	−0.109	−0.067
NO2	0.142	−0.241	−0.072
PO4	0.220	0.333	−0.377
pH	0.344	0.156	−0.221
Dissolved oxygen (DO)	−0.290	0.179	0.418
Electric conductivity (EC)	0.541	−0.114	0.138
Turbidity (TUR)	0.133	−0.090	0.564
Water temperature (TEM)	0.342	0.398	0.039
Current velocity (VEL)	0.208	0.316	0.482
Average depth (DEP)	0.047	0.334	−0.081
Average width (WID)	−0.088	0.530	0.089

Note: Acronyms of variables used in PCA and NMDS ordinations are in parenthesis.

3.2 | Macroinvertebrate Composition

A total of 6507 macroinvertebrates were collected, which were classified into 47 genera from 33 families and 12 orders (Table S1), 29 of which were recorded for the first time in streams located in the western region of Ecuador (Table S1). Seventy-nine percent of the total number of macroinvertebrates collected belonged to three orders of insects: hemipterans (60%, with *Rhagovelia* and *Husseyella* being the most abundant genera), ephemeropterans (11%, *Thraulodes*) and coleopterans (8%, *Laccodytes*). The most abundant non-insect was the amphipod *Hyalella*, which constituted 11% of the total number of macroinvertebrates collected (Table S1).

In global terms, there was greater taxon richness in summer (46 genera) than in winter (40 genera), with the hemipterans *Rhagovelia* and *Husseyella* being the most abundant taxa in summer, and *Rhagovelia* and the amphipod *Hyalella* being the most abundant in winter (Table S1).

With regard to the contribution made by invertebrate orders to streams, the most dominant group comprised hemipterans (Figure 3a). The highest proportion of hemipterans and amphipods was found in Santa Rosa, whereas the highest proportion of ephemeropterans was in Pozo Hondo, and the highest proportion of coleopterans was in Buenos Aires. Regarding the seasons, the populations of odonates, hemipterans, and coleopterans decreased in winter, whereas those of amphipods, trichopterans, ephemeropterans, and dipterans increased (Figure 3a).

3.3 | Macroinvertebrate Functional Feeding Groups

Of the total number of macroinvertebrates collected, 94% comprised three FFGs: predators (69%, with *Rhagovelia* and *Husseyella*

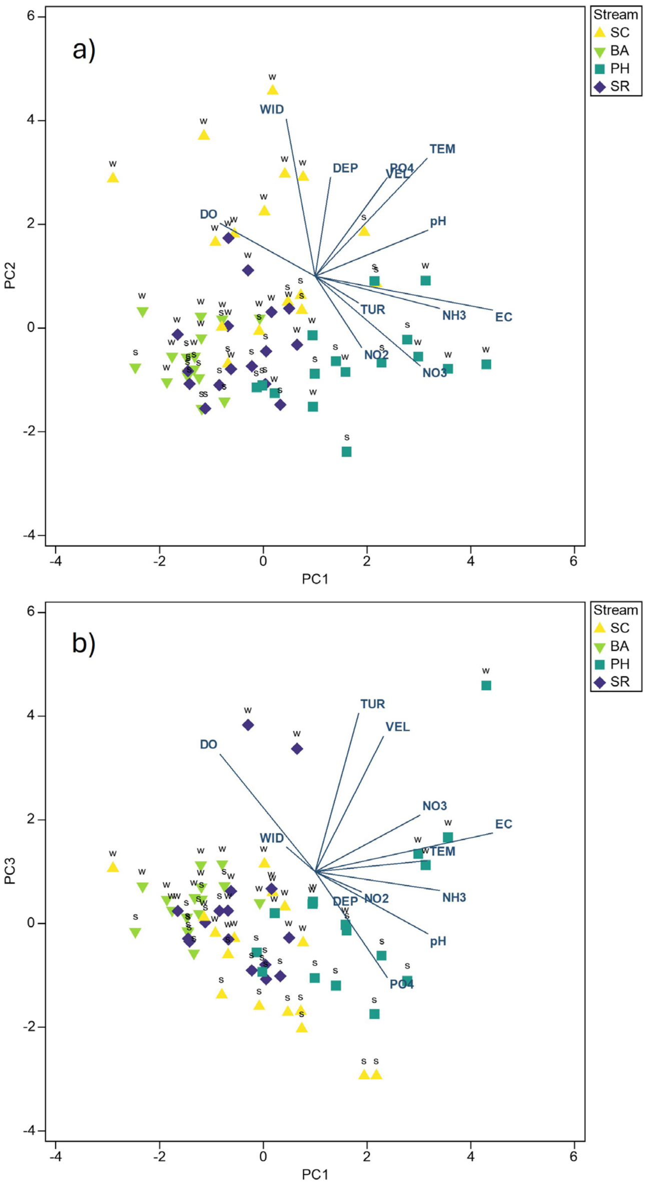

FIGURE 2 | Principal Component Analysis (PCA) performed on the environmental (abiotic) matrix. (a) First component (axis) versus second component (PC1—PC2 ordination). (b) First component (axis) versus. third component (PC1—PC3 ordination). Samples are additionally labeled by seasons: Summer samples (s) and winter samples (w). San Carlos (SC), Buenos Aires (BA), Pozo Hondo (PH), and Santa Rosa (SR). Acronyms of environmental variables as in Table 2. [Color figure can be viewed at wileyonlinelibrary.com]

being the most abundant genera), shredders (13%, *Hyalella*), and collector-gatherers (12%, *Thraulodes* and *Leptohyphes*) (Table S1).

With regard to the contribution made by FFGs to streams, predators were the dominant group (Figure 3b). Buenos Aires contained the highest proportion of predators, whereas Pozo Hondo had the highest proportion of collector-gatherers and collector-filterers. Moreover, San Carlos had the highest proportion of scrapers, whereas Santa Rosa contained the highest proportion of shredders. With regard to the seasons, the populations of predators and scrapers decreased in winter, whereas those of shredders and collectors increased (Figure 3b).

3.4 | Univariate Statistical Analysis of Community Metrics

The mean values of the community metrics (richness, abundance, and diversity) obtained for streams and seasons are shown in Table 3. The significant differences between streams and seasons in benthic macroinvertebrate diversity, abundance, and richness were analyzed by performing linear models (Table 4). In the case of macroinvertebrate diversity (Model 1), the best model found significant differences between streams and seasons: Santa Rosa stream obtained the lowest diversity ($\beta = -0.21$, Table 4; Figure 4b), and its summer communities were more diverse than its winter communities ($\beta = -0.14$; Figure 4d). With regard to macroinvertebrate abundance (Model 2), the best model also showed significant differences between streams and seasons, with the lowest values being obtained for Pozo Hondo stream ($\beta = -1.52$; Table 4; Figure 4a), and a higher abundance in summer than in winter for all the streams ($\beta = -0.43$; Figure 4c). Finally, there was a significant difference in taxon richness (Model 3) according to the season, with higher values in summer than in winter ($\beta = -0.52$; Figure 4e), but there were no significant differences between streams (and, therefore, data not shown in Figure 4).

3.5 | Multivariate Analysis of Macroinvertebrate Assemblages

The PERMANOVA analysis performed on abundance data of macroinvertebrate genera confirmed significant differences in the structure of macroinvertebrate assemblages across streams ($p = 0.001$) and seasons ($p = 0.001$) (Table 5). A significant interaction between stream and season was found ($p = 0.004$), indicating that the difference amongst streams regarding macroinvertebrate structure was dependent on seasonal variability. The NMDS biotic ordination of samples performed on the abundance data of macroinvertebrate genera visualized the differences found in the PERMANOVA analysis between streams and seasons (Figure 5a). Environmental vectors (Pearson correlation coefficients between environmental variables and the sample scores for NMDS1 and NMDS2 axes) were overlaid on the NMDS plot (Table S2). According to this analysis, pH, phosphate, and DO were the main variables to significantly correlate with NMDS1, representing a eutrophication gradient caused principally by phosphates. The summer samples from the San Carlos and Buenos Aires streams, where the highest values of phosphates and pH were recorded, were located on the positive side. Conversely, the high concentrations of DO, recorded in the winter samples from San Carlos, Pozo Hondo, and Buenos Aires, were located on the negative side. NMDS2 was correlated principally with average width, water temperature, and average depth, representing a stream size gradient. The winter samples from the second-order stream San Carlos, affected by the increased discharge due to rainfall, were located on the positive side, apart from the first-order streams. The summer samples from the smallest streams (Buenos Aires and Santa Rosa) were located at the negative-down position, reflecting the decrease in discharge (Figure 5a). The differences among the streams, therefore, depended on seasonal changes, as demonstrated by the significant PERMANOVA interaction term *stream × season* (Table 5).

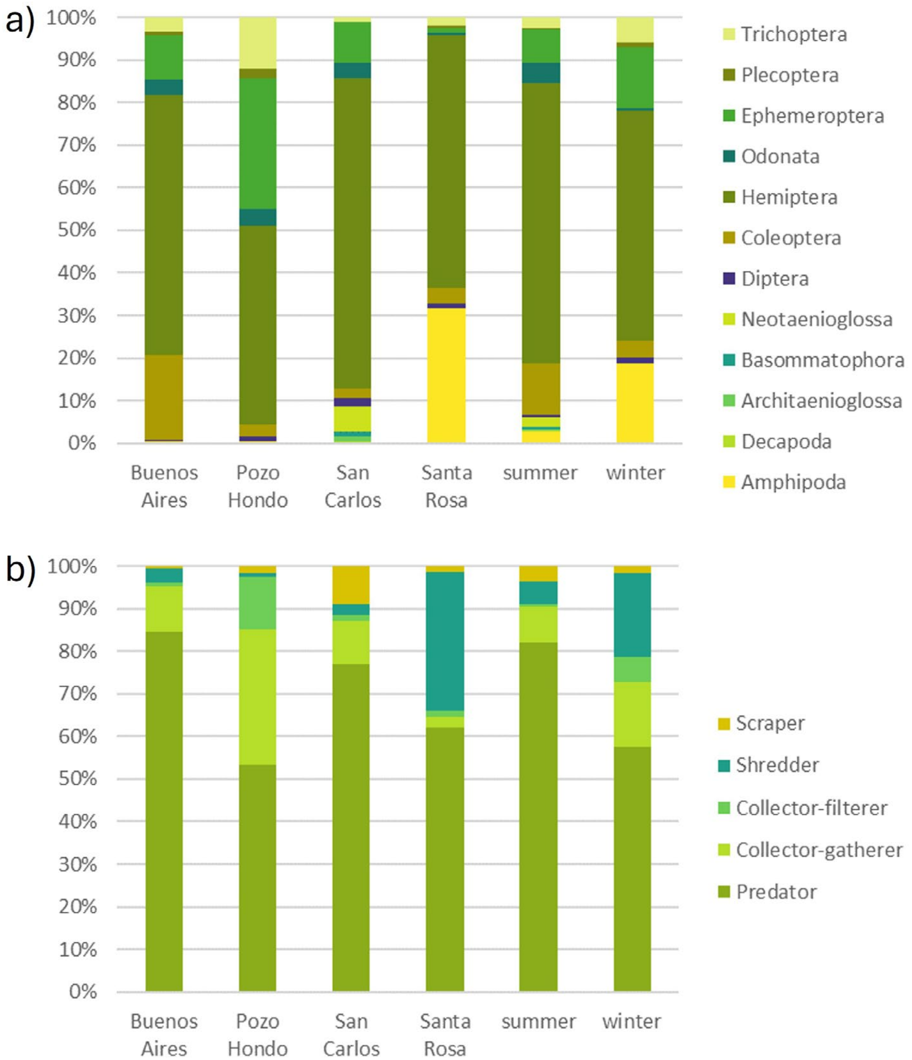

FIGURE 3 | (a) Invertebrate composition (% of total captures) in streams and seasons. (b) Functional feeding groups (% of total captures) in streams and seasons. [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 3 | Mean values and standard deviation (sd) of community metrics for the streams and seasons studied.

	Richness		Abundance		Diversity	
	Mean	**sd**	**Mean**	**sd**	**Mean**	**sd**
Stream						
Buenos Aires (BA)	11.25	3.89	115.25	66.23	0.71	0.18
Pozo Hondo (PH)	10.94	3.43	79.75	68.51	0.79	0.18
San Carlos (SC)	9.56	5.20	73.06	76.12	0.70	0.23
Santa Rosa (SR)	8.81	4.07	138.63	106.90	0.51	0.13
Stream average	10.14	4.15	101.67	79.44	0.68	0.18
Season						
Summer (s)	10.88	4.52	98.78	64.56	0.72	0.23
Winter (w)	9.41	3.83	104.56	100.06	0.64	0.18

TABLE 4 | Results of linear models. *F* values and coefficients for the levels of factors stream and season included in the best models to explain benthic macroinvertebrate diversity (Model 1), abundance of individuals (Model 2) and taxon richness (Model 3).

Variables	*F*	Coefficient ± SE
Diversity (Model 1)		
Stream	**6.34*****	Pozo Hondo = 0.05 ± 0.07
		San Carlos = −0.03 ± 0.06
		Santa Rosa = −0.21 ± 0.06
Season	**6.44***	Winter = −0.14 ± 0.05
Abundance (Model 2)		
Stream	**90.55*****	Pozo Hondo = −1.52 ± 0.11
		San Carlos = 0.02 ± 0.06
		Santa Rosa = −0.08 ± 0.04
Season	**88.27*****	Winter = −0.43 ± 0.05
Richness (Model 3)		
Season	**22.03*****	Winter = −0.52 ± 0.11

Note: Coefficients for the level of fixed factors were calculated using the reference value of "Buenos Aires" in the variable "Stream" and "Summer" in the variable "Season." Significant effects are shown in bold type: *$p < 0.05$, ***$p < 0.001$.

3.6 | Matching Macroinvertebrate Assemblages to Abiotic Conditions

The alignment between the biotic NMDS ordination and the results obtained from the abiotic ordination (PCA) is noteworthy (Figure 2). The environmental vectors overlaid on the biological NMDS ordination (Figure 5a) had nearly the same direction and relative contributions as those in the PCA ordination. Therefore, it is possible to interpret that the macroinvertebrate assemblages responded well to the environmental gradients defined in the PCA. Stream width had the highest Pearson correlation coefficients with NMDS2 ($r = 0.538$; Table S2) and, simultaneously, the highest gradient with PC2 (eigenvector = 0.53; Table 2), placing the winter samples collected from San Carlos in the upper negative side in both ordinations. The summer samples were predominantly located on the negative side of the NMDS2 axis, a trend that resembles the seasonality gradient of PC3. Conductivity, the main contributor to PC1 (Table 2), had not a significant influence the ordination of the samples, since the correlation values with the NMDS axes were very weak (Table S2). However, pH ($r = 0.492$; Table S2), phosphate ($r = 0.490$), and DO ($r = -0.364$) significantly correlated with NMDS1, whereas pH made a significant contribution to the PC1 axis in the abiotic ordination (Table 2). Finally, DO contributed significantly to PC3, while phosphate contributed to both PC2 and PC3 axes (Table 2).

The agreement between the abiotic and biotic ordinations motivated the analysis of the environmental variables that best explained resemblances among assemblages (BEST routine, PRIMER v7 software). The maximum correlation value was $\rho = 0.384$, attained by the combination of two environmental variables: phosphate concentration (PO4) and stream width (WID) (Table S3). Note that the same maximum ρ value was also reached with the combination of five environmental variables: phosphate, DO, water temperature, current velocity, and stream width (Table S3). Therefore, these five variables were the main environmental drivers that best explained macroinvertebrate assemblage patterns. A global test performed on permutational analyses (99 permutations of random samples) indicated that the BEST statistic $\rho = 0.384$ was significant ($p = 0.001$; Table S3). According to these results, the macroinvertebrate samples taken from the second-order stream San Carlos (the widest stream and the stream with the highest phosphate concentration; Table 1) were the most dissimilar from the other three first-order streams as it can be observed in the biotic NMDS (Figure 5a). Furthermore, seasonality had an additional influence on these patterns, since the greatest width was recorded for San Carlos in winter, and the highest phosphate concentration was recorded for the same stream in summer, a pattern that can be visualized in the NMDS with overlaid environmental vectors (Figure 5a). This result is consistent with the PERMANOVA analysis, in which the interaction term *stream* × *season* had a significant effect on the abundance of macroinvertebrate assemblages.

Finally, the genera vectors were overlaid on the NMDS ordination to identify the macroinvertebrate taxa that contributed most to the differences in assemblages (Figure 5b; Table S4). The genera vectors are oriented to the samples with its highest abundance. The highest Pearson correlation coefficients on the positive side of NMDS1 ($r > 0.4$; Table S4) were obtained by the hemipterans *Eurygerris* and *Pelocoris*, the mollusk *Pomacea* and the odonate *Acanthagrion*, corresponding to summer samples collected from the San Carlos stream (Figure 5b). The heteropterans *Rhagovelia* and *Husseyella* were located on the most negative side of NMDS1 (Table S4), with their highest abundances recorded for summer samples taken from the Santa Rosa stream (Table S1). The winter samples collected from the San Carlos stream were located on the most positive side of NMDS2, with the ephemeropteran *Paracloeodes* and the hemipterans *Hydrometra* and *Halobatopsis* obtaining the highest positive correlation values ($r > 0.39$; Table S4). Finally, the summer samples taken from the Buenos Aires stream were located on the most negative side of NMDS2, with the trichopteran *Phylloicus*, the heteropteran *Rhagovelia*, and the coleopteran *Laccodytes*, obtaining the highest negative correlation values (Table S4; Figure 5b).

4 | Discussion

4.1 | Macroinvertebrate Composition

Of the registered 47 macroinvertebrate genera, 29 were recorded for the first time in streams located in the western region of Ecuador (Table S1). Other genera have been recorded previously in Ecuadorian regions, for example, five genera (*Leptohyphes, Tricorythodes, Limnocoris, Heterelmis*, and *Cylloepus*) were found in streams in the Andes mountains in northern Ecuador (Sites, Willig, and Linit 2003); eight genera (*Anchytarsus, Hexatoma, Leptohyphes, Thraulodes, Hetaerina, Phylloicus, Leptonema*, and *Limnocoris*) were found in cloud forest streams in southern Ecuador (Bücker et al. 2010); and seven genera (*Thraulodes,*

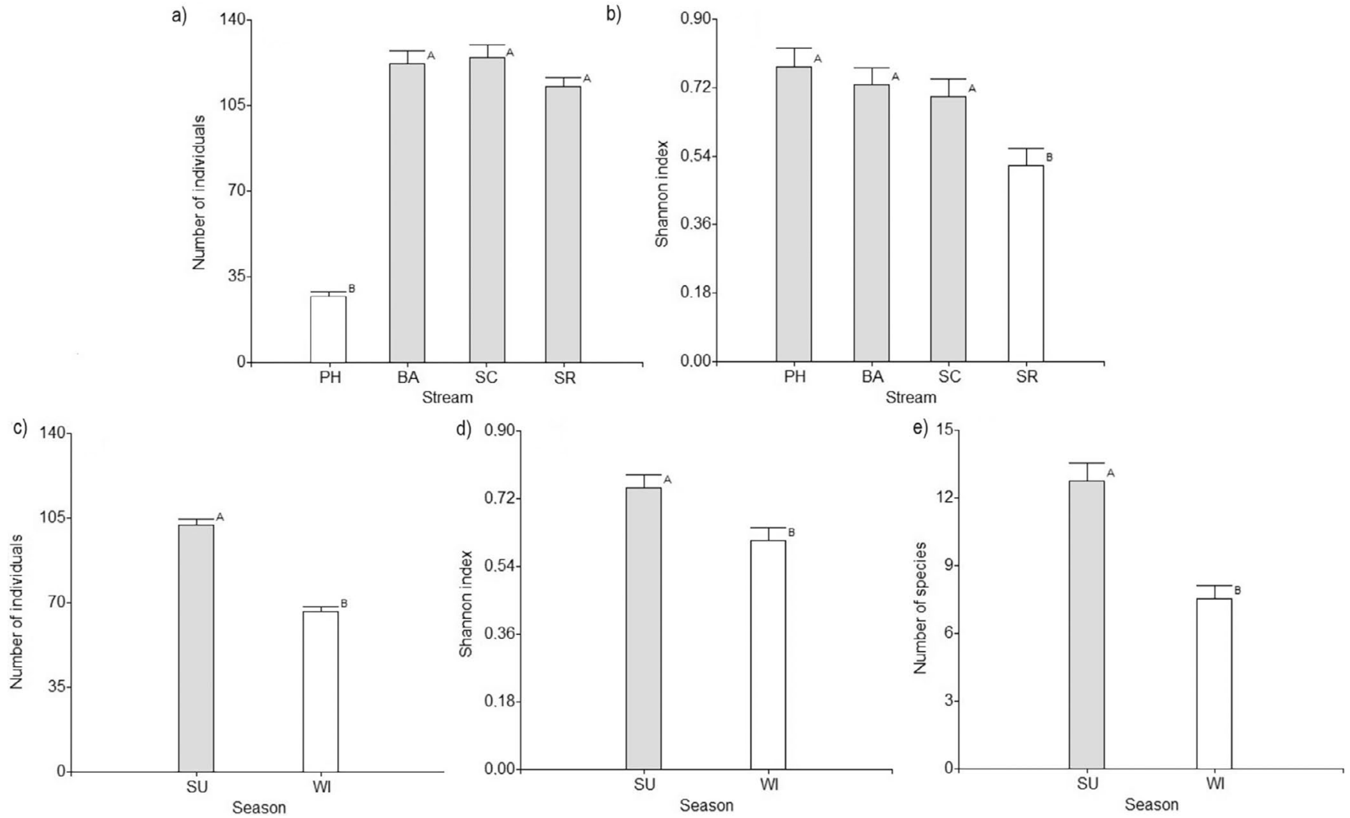

FIGURE 4 | Predicted mean values of macroinvertebrate abundance (a) and diversity (b) by streams. Macroinvertebrate abundance (c), diversity (d) and richness (e) by seasons. Whiskers represent the standard error.

TABLE 5 | Permutational Multivariate Analysis of Variance (PERMANOVA) of macroinvertebrate assemblages (abundance data) based on the factors stream (St) and season (Se, nested in year, Ye).

Source	*df*	SS	MS	Pseudo-*F*	*p*	Unique permutations
St	3	40,330	13443.0	7.8789	0.001	998
Se (Ye)	1	8109.1	8109.1	4.7527	0.001	999
St × Se (Ye)	3	9888.7	3296.2	1.9319	0.004	997
Residual	56	95,549	1706.2			
Total	63	1.54E+05				

Leptohyphes, Tricorythodes, Microvelia, Rhagovelia, Phylloicus, and *Leptonema*) were found in streams located in the northern part of the Ecuadorian coast (Martínez-Sanz et al. 2014).

4.2 | Functional Feeding Groups

In general, the patterns of FFG composition observed in our study area can be summarized as follows: (i) predators (mainly heteropterans) were generally dominant in all the streams, (ii) shredders (amphipods) were the second most abundant group, although they were found almost exclusively in the Santa Rosa stream; (iii) collector-gatherers (ephemeropterans) were well distributed in the four streams, and (iv) scrapers (mollusks) were virtually absent, attaining the greatest abundance in the San Carlos stream.

Several environmental characteristics of the streams may explain the dominance of FFGs. For instance, Pozo Hondo, characterized by higher mineralization and eutrophication, along with the high percentage of agricultural land in its catchment, could explain the dominance of collector-gatherers and collector-filterers, which are usually found in streams subject to high levels of human disturbance (Edegbene et al. 2022; Lima et al. 2022; Malacarne, Machado, and Moretto 2023). The largest stream was San Carlos (the only second-order stream), with the highest channel width and depth, and it recorded the highest phosphate values. The wider channel of San Carlos allowed light to penetrate the streambed, the rocks on which were covered by a considerable abundance of algae. This factor, combined with high phosphate levels, probably stimulated algal growth, thus explaining the dominance of scrapers in the San Carlos stream (Cummins and Klug 1979). A similar finding was reported by

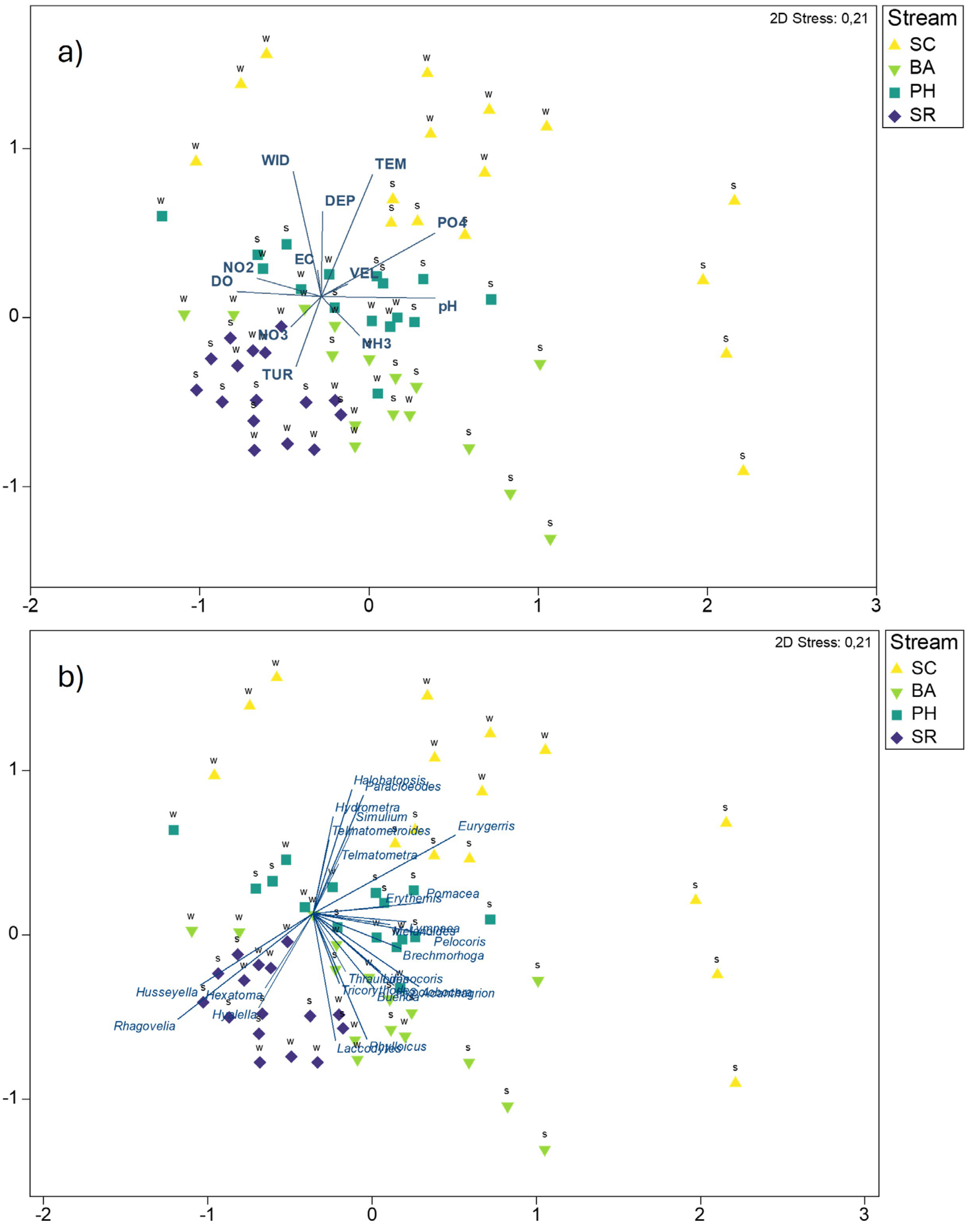

FIGURE 5 | Nonmetric multidimensional scaling (NMDS) ordination of macroinvertebrate samples collected from the four streams studied: San Carlos (SC), Buenos Aires (BA), Pozo Hondo (PH), and Santa Rosa (SR). Samples are additionally labeled by seasons: Summer samples (s) and winter samples (w). (a) Environmental vectors (acronyms as in Table 2) are overlaid on biological ordination; vector arrows indicate the direction of the increasing environmental gradient, and vector lengths are proportional to their correlation with NMDS axes (Table S2).(b) Taxa vectors are overlaid on the biological ordination and represent the correlation between taxa abundances and NMDS axes (Table S4). [Color figure can be viewed at wileyonlinelibrary.com]

Cortés-Guzmán, Alcocer, and Cummins (2021) for a low-shade tropical stream, where a high biomass of scrapers was reported in association with benthic algae.

The River Continuum Concept (RCC) proposed by Vannote et al. (1980) predicts the dominance of shredders and collectors in headwater low-order streams. However, there was a dominance of predators, mainly pleustonic heteropterans (Veliidae, Gerridae), in the headwater streams studied herein. These predators feed primarily on terrestrial invertebrates that fall onto the surface of the water from the dense riparian forest (Nakano and Murakami 2001; Zhang and Richardson 2011).

According to Kohlmann et al. (2021), the functional group assemblage in tropical forests differs from that hypothesized by Vannote et al. (1980) for temperate rivers, since predators appear to play a much more critical role than shredders in tropical streams. Other studies on neotropical rivers have also reported this greater significance of predators when compared to shredders (Van Echelpoel et al. 2018). This was the case of the Buenos Aires and Santa Rosa streams, both of which had a denser canopy that favored the presence of predators. A similar result was also attained by Masese et al. (2014) for Kenyan highland streams. Moreover, the Santa Rosa and Buenos Aires streams had more pools than riffles, a preferred habitat for predators such as Hemiptera (Galeti, Capitanio, and Baldissera 2020).

The second most abundant FFG in the streams studied was shredders, owing to the high input of litter from the riparian forest. The Santa Rosa stream had the highest abundance of shredders (the amphipod *Hyalella*), which could also be favored by its fastest flow and lowest temperature among the four studied streams. According to Rawi et al. (2014), shredders in tropical forest streams in Malaysia are positively correlated with the current velocity, and Amphipoda in particular usually prefer habitats in rapid currents (e.g., Correa-Araneda and De Los Ríos 2010). Moreover, the presence of shredders in rivers of the Ecuadorian Amazon has been potentially associated with decreases in temperature (Cabrera et al. 2021).

In terms of temporal variation, the populations of predators and scrapers decreased during winter, while those of shredders and collectors increased. The pattern observed for the composition of the FFGs of our macroinvertebrate communities, in which predators predominated in all the streams studied, particularly in summer, closely resembled the findings of Van Echelpoel et al. (2018) and Cabrera et al. (2021). These authors suggested that the high relative abundances and distributions of predators may be linked to seasonal variations and the absence of certain anthropogenic disturbances, resulting in a greater diversity and number of predators during the dry season when compared to the rainy season. Furthermore, the rapid turnover rates of prey biomass in summer could lead to the dominance of predators (Cortés-Guzmán, Alcocer, and Cummins 2021), while the increase in collectors during winter could be related to the availability of food resources such as fine particulate organic matter (FPOM) at that time (Cortés-Guzmán, Alcocer, and Cummins 2021).

The increase in shredders during the winter could be related to the food resources that are available. According to Cummins and Klug (1979) and Vannote et al. (1980), shredders in temperate headwater streams typically prefer coarse particulate organic matter (CPOM), such as the allochthonous leaf litter generated in autumn-winter. However, in tropical streams, leaf fall occurs during summer, making CPOM potentially more important in summer than in winter. We hypothesize that CPOM and FPOM may be mobilized during winter due to the increased frequency of rainstorms (the rainy season), leading to higher discharge, water velocity and turbidity. Under these conditions, amphipods could feed on mobilized FPOM and CPOM, resulting in increased populations. In other neotropical streams, it is suggested that shredders might prefer FPOM to CPOM as food resources (Tomanova, Goitia, and Helešic 2006).

4.3 | Community Metrics: Abundance, Richness, and Diversity

The values of richness and diversity (Table 3) in the streams studied herein were low compared to, for example, the values registered for other coastal streams in northern Ecuador (Martínez-Sanz et al. 2014). The aforementioned authors recorded 70 taxa at genera/family level in the whole study area (vs. our 47 genera; Table S1). The richness values for individual streams ranged from 14 to 32 taxa, and the Shannon diversity values ranged between 2 and 4 (vs. our average of 10 genera per stream and diversity values <1; Table S1, Table 3). The larger size of these northern streams (4–9 m width and up to 0.9 m depth) could explain the greater richness of taxa in comparison with that in our small headwater streams (an average of <1 m width and a few centimeters depth). In general, more species are found in large rivers than small streams, mainly because the spatial area and habitat diversity are greater in large systems (Allan 1995). In addition, those streams in the mentioned study were at lower altitudes (25–195 m.a.s.l) than our mountain headwater streams (426–647 m.a.s.l). According to Jacobsen (2003, 2004), some measures of stream diversity correlate significantly with altitude; furthermore, the effect of altitude explained more of the total variability in terms of the richness of invertebrates. Similar results were reported by Castro et al. (2019) in headwater streams along an altitudinal gradient in the Espinhaço mountains in southeastern Brazil, and the same occurred with streams along an altitudinal gradient in southeastern Brazil in a study by Henriques-Oliveira and Nessimian (2010).

The predicted mean values of diversity and richness were significantly lower in winter than in summer (Figure 4d and Figure 4e). A similar trend was reported by Urdanigo et al. (2019) in a creek located in the central coastal region of Ecuador, with a decrease in EPT taxa (*Ephemeroptera*, *Plecoptera*, and *Trichoptera*) during the rainy season. These authors stated that higher precipitation rates significantly increased creek flow, with the consequent drift effect on individuals and the removal of insects. In our study, there was an increase in discharge in winter as the result of rainfall, which also caused a hydrological disturbance that decreased the richness and diversity values and changed the community structure (see following subsection).

The predicted mean values of abundance (Figure 4a) for macroinvertebrate genera were low in all the streams studied compared to other coastal headwater streams in Ecuador (Martínez-Sanz et al. 2014; Urdanigo et al. 2019). The small size of our mountain headwater streams could again be the best explanation. Moreover, the predicted mean values of macroinvertebrates abundance were higher in summer than in winter (Figure 4c). This difference is attributed to lower flow conditions and greater environmental stability during the summer, which allowed macroinvertebrate populations to increase. Similar results were reported by Ramírez, Pringle, and Douglas (2006) for six lowland streams in Costa Rica, by Ríos-Touma, Encalada, and Prat Fornells (2011) for an Ecuadorian highland stream, and by Serna, Tamaris-Turizo, and Gutiérrez Moreno (2015) for four lowland streams in the Colombian Caribbean.

4.4 | Environmental Factors

The biotic-abiotic matching analysis (BEST routine) selected five variables as the environmental factors that best explained assemblage structure: phosphate concentration and stream width, along with DO, water temperature, and current velocity. The NMDS biotic ordination also showed high significant correlation values with these environmental variables (except for current velocity). The highest value of phosphate found in the San Carlos stream could be related to the agricultural activities that take place in summer, since urban wastewaters were absent. Some other neotropical studies have reported similar results: for example, phosphates associated with water pollution affected community composition in a tropical Panamanian stream (Sánchez-Argüello et al. 2010). Moreover, water temperature had a negative effect on dipterans in the rivers in the Eastern Andean Hills of Bogotá (Buitrago-Guacaneme et al. 2018). The influence of hydromorphological characteristics, such as the increase in the current velocity and the average width in winter was associated with the decrease in macroinvertebrate diversity, abundance, and richness. This pattern is linked to an increase in discharge and the impact of hydrological disturbance. A similar result was reported by Nguyen et al. (2018), who stated that stream current velocity was one of the major environmental variables determining the distribution of macroinvertebrate communities in the Guayas River basin (Ecuador). A typical condition of tropical rivers is the decrease in abundance of macroinvertebrate communities during the months with greater rainfall as a result of physicochemical and flow variations (Quesada-Alvarado et al. 2020). In general, many studies conducted in Neotropical streams have recorded these changes in macroinvertebrate abundance and composition linked to seasonal rainfall patterns (Jacobsen and Encalada 1998; Serna, Tamaris-Turizo, and Gutiérrez Moreno 2015; Hernandez-Abrams et al. 2023). It would, therefore, appear that physical variables are important as regards structuring macroinvertebrate communities in neotropical headwater streams (e.g., Ferreira et al. 2014).

As the PERMANOVA analysis showed, the interaction term *stream × season* had a significant effect, indicating that the differences among the streams were dependent on seasonal variability. This effect was clearly reflected in the differences between the assemblages found in the San Carlos second-order stream and those in the remaining first-order streams with respect to the main environmental drivers (phosphate and stream width): the increase in phosphates in the San Carlos stream occurred in summer, while the increase in the width of the San Carlos stream occurred in winter. This result was also good agreement with the PCA analysis and the NMDS-environmental vector plot.

5 | Conclusions

Seasonality plays a critical role in shaping macroinvertebrate assemblages in headwater coastal streams. Winter rainfall significantly increases discharge, causing declines in abundance, diversity, and richness, which in turn alter assemblage composition and trophic structures. The high number of newly recorded genera indicates that this tropical region remains understudied, and further research is essential to better understand the implications of seasonal variability on freshwater biodiversity and ecosystem functioning in neotropical streams. Predators, particularly pleustonic hemipterans, were the most abundant and widely distributed macroinvertebrates alongside collector-gatherers, while mollusks and crustaceans were confined to single streams. Environmental drivers such as trophic condition (phosphorus), stream size, DO, water temperature, and current velocity significantly influenced macroinvertebrate communities. The selected streams represented a wide range of environmental and biotic conditions, highlighting the high diversity of stream types in this mountainous coastal region and the need for extensive future studies to capture the full range of aquatic ecosystems. These findings underscore the intricate relationship between hydrological dynamics and biotic communities, emphasizing the importance of studying these interactions to support effective conservation strategies.

Acknowledgments

We would like to thank S. Newton for reviewing the English used in the manuscript. The authors would additionally like to thank the Biology Laboratory (ECOTEC University), the Aquatic Ecosystems Research Laboratory (ESPOL University), and the Bromatology Laboratory (UNESUM University) staff for their assistance during the work reported in this paper. The water quality analysis and the identification of macroinvertebrates were conducted in the Biology Laboratory (ECOTEC University). Special thanks to Carlos Parrales for his support during the field work, and to Michelle Arrobo for her support during the laboratory work.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data will be available on Supporting Information.

References

Allan, J. D. 1995. *Stream Ecology. Structure and Function of Running Waters*. Kluwer Academic Press.

Anderson, M. 2008. *PERMANOVA+ for PRIMER: Guide to Software and Statistical Methods*. Primer-E Limited.

Barboza, A. C. G., V. Tagliacollo, and G. B. Jacobucci. 2024. "Influence of Seasonal Hydrological Regimes on Benthic Macroinvertebrates in Two the Brazilian Biodiversity Hotspots." *Limnologica* 106: 126170.

Bücker, A., M. Sondermann, H. G. Frede, and L. Breuer. 2010. "The Influence of Land-Use on Macroinvertebrate Communities in Montane Tropical Streams-a Case Study From Ecuador." *Fundamental and Applied Limnology* 177, no. 4: 267–282.

Buitrago-Guacaneme, A., A. Sotelo-Londoño, G. A. Pinilla-Agudelo, A. García-García, L. I. Moneada, and P. H. Adler. 2018. "Abundance and Diversity of Black Flies (Diptera: Simuliidae) in Rivers of the Andean Eastern Hills of Bogotá (Colombia), and Its Relationship With Water Stream Physicochemical Variables." *Universitas Scientiarum* 23, no. 2: 291–317.

Cabrera, S., M. A. Eurie Forio, K. Lock, et al. 2021. "Variations in Benthic Macroinvertebrate Communities and Biological Quality in the Aguarico and Coca River Basins in the Ecuadorian Amazon." *Water* 13, no. 12: 1692.

Cabrera-García, S., P. L. Goethals, K. Lock, et al. 2023. "Taxonomic and Feeding Trait-Based Analysis of Macroinvertebrates in the Antisana River Basin (Ecuadorian Andean Region)." *Biology* 12, no. 11: 1386.

Castro, D. M., M. Callisto, R. R. Solar, D. R. Macedo, and G. W. Fernandes. 2019. "Beta Diversity of Aquatic Invertebrates Increases Along an Altitudinal Gradient in a Neotropical Mountain." *Biotropica* 51, no. 3: 399–411.

Chi, S., S. Li, S. Chen, M. Chen, J. Zheng, and J. Hu. 2017. "Temporal Variations in Macroinvertebrate Communities From the Tributaries in the Three Gorges Reservoir Catchment, China." *Revista Chilena de Historia Natural* 90: 1–11.

Clarke, K. R., and R. N. Gorley. 2006. *PRIMER v6: User Manual/ Tutorial*, 78. Primer-E.

Correa-Araneda, F., and P. De Los Ríos. 2010. "Amphipoda and Decapoda as Potential Bioindicators of Water Quality in an Urban Stream (38 S, Temuco, Chile)." *Crustaceana* 83: 897–902.

Cortés-Guzmán, D., J. Alcocer, and K. W. Cummins. 2021. "Benthic Macroinvertebrates of Tropical Streams: Functional and Trophic Diversity of the Lacantún River, Mexico." *Limnology* 22, no. 3: 313–328.

Cortés-Guzmán, D., J. Alcocer, and D. Planas. 2022. "Autotrophs Are Important Contributors to Benthic Macroinvertebrate Food Webs in Two Tropical First-Order Forest Streams." *Freshwater Biology* 67, no. 6: 941–953.

Couceiro, S. R., K. Dias-Silva, and N. Hamada. 2021. "Influence of Climate Seasonality on the Effectiveness of the Use of Aquatic Macroinvertebrates in Urban Impact Evaluation in Central Amazonia." *Limnology* 22, no. 2: 237–244.

Crespo-Pérez, V., P. Andino, R. Espinosa, O. Dangles, and D. Jacobsen. 2016. "The Altitudinal Limit of Leptohyphes Eaton, 1882 and Lachlania Hagen, 1868 (Ephemeroptera: Leptohyphidae, Oligoneuriidae) in Ecuadorian Andes Streams: Searching for Mechanisms." *Aquatic Insects* 37, no. 1: 69–86.

Crespo-Pérez, V., E. Kazakou, D. W. Roubik, and R. E. Cárdenas. 2020. "The Importance of Insects on Land and in Water: A Tropical View." *Current Opinion in Insect Science* 40: 31–38.

Cummins, K. W., and M. J. Klug. 1979. "Feeding Ecology of Stream Invertebrates. Annual Review of Ecology and Systematics." *Annual Review of Ecology and Systematics* 10, no. 1: 147–172.

Da Silva Giehl, N. F., P. V. Fonseca, K. Dias-Silva, T. P. Mendes, and H. S. R. Cabette. 2022. "Environmental and Seasonal Effects on Gerridae Assemblages (Heteroptera: Gerromorpha) From a Stream in Brazilian Savannah." *Neotropical Entomology* 51, no. 3: 376–385.

Damanik-Ambarita, M. N., K. Lock, P. Boets, et al. 2016. "Ecological Water Quality Analysis of the Guayas River Basin (Ecuador) Based on Macroinvertebrates Indices." *Limnologica* 57: 27–59.

Di Rienzo, J. A., F. Casanoves, M. G. Balzarini, L. González, M. Tablada, and Y. C. Robledo. 2011. "InfoStat Version 2011." Grupo InfoStat, FCA, Universidad Nacional de Córdoba, Argentina. http://www.infostat.com.ar.

Domínguez, E., and H. R. Fernández. 2009. *Macroinvertebrados bentónicos sudamericanos. Sistemática y Biología*, 656. Fundación Miguel Lillo.

Dominguez-Granda, L., K. Lock, and P. L. Goethals. 2011. "Using Multi-Target Clustering Trees as a Tool to Predict Biological Water Quality Indices Based on Benthic Macroinvertebrates and Environmental Parameters in the Chaguana Watershed (Ecuador)." *Ecological Informatics* 6, no. 5: 303–308.

Van Echelpoel, W., M. A. E. Forio, J. Van Butsel, et al. 2018. "Macroinvertebrate Functional Feeding Group Structure Along an Impacted Tropical River: The Portoviejo River (Ecuador)." *Limnologica* 73: 12–19.

Edegbene, A. O., F. C. Akamagwuna, F. O. Arimoro, E. C. Akumabor, and E. A. Kaine. 2022. "Effects of Urban-Agricultural Land-Use on Afrotropical Macroinvertebrate Functional Feeding Groups in Selected Rivers in The Niger Delta Region, Nigeria." *Hydrobiologia* 849, no. 21: 4857–4869.

Escribano-Ávila, G. 2016. "El Bosque Seco Neotropical de la Provincia Ecuatoriana: Un pequeño gran desconocido." *Ecosistemas: Revista Cietifica y Tecnica de Ecologia y Medio Ambiente* 25, no. 2: 1–4.

Ferreira, W. R., R. Ligeiro, D. R. Macedo, et al. 2014. "Importance of Environmental Factors for the Richness and Distribution of Benthic Macroinvertebrates in Tropical Headwater Streams." *Freshwater Science* 33, no. 3: 860–871.

Gabriels, W., K. Lock, N. De Pauw, and P. L. Goethals. 2010. "Multimetric Macroinvertebrate Index Flanders (MMIF) for Biological Assessment of Rivers and Lakes in Flanders (Belgium)." *Limnologica-Ecology and Management of Inland Waters* 40, no. 3: 199–207.

Galarza, E., M. Cabrera, R. Espinosa, E. Espitia, G. M. Moulatlet, and M. V. Capparelli. 2021. "Assessing the Quality of Amazon Aquatic Ecosystems With Multiple Lines of Evidence: The Case of the Northeast Andean Foothills of Ecuador." *Bulletin of Environmental Contamination and Toxicology* 107: 52–61.

Galeti, G., B. M. Capitanio, and R. Baldissera. 2020. "Variation of Benthic Macroinvertebrate Communities in Streams of Three Landscapes of South Brazilian Grasslands." *Revista de Biología Tropical* 68, no. 1: 108–121.

Hamada, N., J. H. Thorp, and D. C. Rogers, eds. 2018. *Thorp and Covich's Freshwater Invertebrates: Keys to Neotropical Hexapoda*. Vol. 3. Academic Press.

Hampel, H., R. F. Vázquez, H. González, and R. Acosta. 2023. "Evaluating the Ecological Status of Fluvial Networks of Tropical Andean Catchments of Ecuador." *Water* 15, no. 9: 1742.

Henriques-Oliveira, A. L., and J. L. Nessimian. 2010. "Aquatic Macroinvertebrate Diversity and Composition in Streams Along an Altitudinal Gradient in Southeastern Brazil." *Biota Neotropica* 10: 115–128.

Hernandez-Abrams, D. D., S. Connelly, M. C. Freeman, P. E. Gutierrez-Fonseca, and S. J. Wenger. 2023. "Seasonal Variability in Macroinvertebrate Assemblages in Paired Perennial and Intermittent Streams in Costa Rica." *Hydrobiologia* 850, no. 1: 215–230.

Hu, X., D. Zuo, Z. Xu, et al. 2022. "Response of Macroinvertebrate Community to Water Quality Factors and Aquatic Ecosystem Health Assessment in a Typical River in Beijing, China." *Environmental Research* 212: 113474.

Jackson, J. K., J. M. Battle, and B. W. Sweeney. 2010. "Monitoring the Health of Large Rivers With Macroinvertebrates: Do Dominant Taxa Help or Hinder the Assessment?" *River Research and Applications* 26, no. 8: 931–947.

Jacobsen, D. 2003. "Altitudinal Changes in Diversity of Macroinvertebrates From Small Streams in the Ecuadorian Andes." *Archiv für Hydrobiologie* 158, no. 2: 145–168.

Jacobsen, D. 2004. "Contrasting Patterns in Local and Zonal Family Richness of Stream Invertebrates Along an Andean Altitudinal Gradient." *Freshwater Biology* 49, no. 10: 1293–1305.

Jacobsen, D., and A. Encalada. 1998. "The Macroinvertebrate Fauna of Ecuadorian Highland Streams in the Wet and Dry Season." *Archiv für Hydrobiologie* 142: 53–70.

Jacobsen, D., R. Schultz, and A. Encalada. 1997. "Structure and Diversity of Stream Invertebrate Assemblages: The Influence of Temperature With Altitude and Latitude." *Freshwater Biology* 38, no. 2: 247–261.

Kohlmann, B., D. Vásquez, A. Arroyo, and M. Springer. 2021. "Taxonomic and Functional Diversity of Aquatic Macroinvertebrate Assemblages and Water Quality in Rivers of the Dry Tropics of Costa Rica." *Frontiers in Environmental Science* 9: 660260.

Kuhn, J., P. Andino, R. Calvez, et al. 2011. "Spatial Variability in Macroinvertebrate Assemblages Along and Among Neighbouring Equatorial Glacier-Fed Streams." *Freshwater Biology* 56, no. 11: 2226–2244.

Lafuente, W., A. J. Carpio, C. Alcácer, and J. L. Moreno. 2023. "Spatio-Temporal Variability of Physicochemical Conditions in the Headwaters of Neotropical Streams." *Journal of South American Earth Sciences* 126: 104361.

Lessmann, J., J. M. Guayasamin, K. L. Casner, et al. 2016. "Freshwater Vertebrate and Invertebrate Diversity Patterns in an Andean-Amazon Basin: Implications for Conservation Efforts." *Neotropical Biodiversity* 2, no. 1: 99–114.

Lim, S. H., and Y. Do. 2023. "Macroinvertebrate Conservation in River Ecosystems: Challenges, Restoration Strategies, and Integrated Management Approaches." *Entomological Research* 53, no. 8: 271–290.

Lima, M., V. C. Firmino, C. K. S. de Paiva, L. Juen, and L. S. Brasil. 2022. "Land Use Changes Disrupt Streams and Affect the Functional Feeding Groups of Aquatic Insects in the Amazon." *Journal of Insect Conservation* 26, no. 2: 137–148.

Malacarne, T. J., N. R. Machado, and Y. Moretto. 2023. "Influence of Land Use on the Structure and Functional Diversity of Aquatic Insects in Neotropical Streams." *Hydrobiologia* 851: 1–16.

Mariani-Ríos, A., N. Maldonado-Benítez, and A. Ramírez. 2022. "Natural History of Odonata Assemblages in Tropical Streams in Puerto Rico." *Neotropical Biodiversity* 8, no. 1: 112–123.

Martínez-Sanz, C., S. M. Puente-García, E. R. Rebolledo, and P. Jiménez-Prado. 2014. "Macroinvertebrate Richness Importance in Coastal Tropical Streams of Esmeraldas (Ecuador) and Its Use and Implications in Environmental Management Procedures." *International Journal of Ecology* 2014: 1–11.

Masese, F. O., N. Kitaka, J. Kipkemboi, G. M. Gettel, K. Irvine, and M. E. McClain. 2014. "Macroinvertebrate Functional Feeding Groups in Kenyan Highland Streams: Evidence for a Diverse Shredder Guild." *Freshwater Science* 33, no. 2: 435–450.

Molinero, J., M. Barrado, M. Guijarro, M. Ortiz, O. Carnicer, and D. Zuazagoitia. 2019. "The Teaone River: A Snapshot of a Tropical River From the Coastal Region of Ecuador." *Limnetica* 38, no. 2: 587–605.

Nakano, S., and M. Murakami. 2001. "Reciprocal Subsidies: Dynamic Interdependence Between Terrestrial and Aquatic Food Webs." *Proceedings of the National Academy of Sciences* 98, no. 1: 166–170.

Nguyen, T. H. T., P. Boets, K. Lock, et al. 2017. "Water Quality Related Macroinvertebrate Community Responses to Environmental Gradients in the Portoviejo River (Ecuador)." In *Annales De Limnologie-International Journal of Limnology*, vol. 53, 203–219. EDP Sciences.

Nguyen, T. H. T., M. A. E. Forio, P. Boets, et al. 2018. "Threshold Responses of Macroinvertebrate Communities to Stream Velocity in Relation to Hydropower Dam: A Case Study From the Guayas River Basin (Ecuador)." *Water* 10, no. 9: 1195.

ODEPLAN (Oficina de Planificación de la Presidencia). 2004. "Plan Nacional de Ordenamiento Territorial y Generación de Información Base del Ecuador." Quito, Ecuador.

Pearson, R. G., F. Christidis, N. M. Connolly, et al. 2017. "Stream Macroinvertebrate Assemblage Uniformity and Drivers in a Tropical Bioregion." *Freshwater Biology* 62, no. 3: 544–558.

Quesada-Alvarado, F., G. Umaña Villalobos, M. Springer, and J. Picado Barboza. 2020. "Variación estacional y características fisicoquímicas e hidrológicas que influyen en los macroinvertebrados acuáticos, en un río tropical." *Revista de Biología Tropical* 68, no. Supl.2: S54–S67.

Ramírez, A., and P. E. Gutiérrez-Fonseca. 2014. "Functional Feeding Groups of Aquatic Insect Families in Latin America: A Critical Analysis and Review of Existing Literature." *Revista de Biología Tropical* 62: 155–167.

Ramírez, A., C. M. Pringle, and M. Douglas. 2006. "Temporal and Spatial Patterns in Stream Physicochemistry and Insect Assemblages in Tropical Lowland Streams." *Journal of the North American Benthological Society* 25, no. 1: 108–125.

Rawi, C. S., S. A. Al-Shami, M. R. Madrus, and A. H. Ahmad. 2014. "Biological and Ecological Diversity of Aquatic Macroinvertebrates in Response to Hydrological and Physicochemical Parameters in Tropical Forest Streams of Gunung Tebu, Malaysia: Implications for Ecohydrological Assessment." *Ecohydrology* 7, no. 2: 496–507.

Reid, A. J., A. K. Carlson, I. F. Creed, et al. 2019. "Emerging Threats and Persistent Conservation Challenges for Freshwater Biodiversity." *Biological Reviews* 94, no. 3: 849–873.

Ríos-Touma, B., F. Cuesta, E. Rázuri-Gonzales, R. Holzenthal, A. Tapia, and M. Calderón-Loor. 2022. "Elevational Biodiversity Gradients in the Neotropics: Perspectives From Freshwater Caddisflies (Insecta: Trichoptera)." *PLoS One* 17, no. 8: e0272229.

Ríos-Touma, B., A. C. Encalada, and N. Prat Fornells. 2011. "Macroinvertebrate Assemblages of an Andean High-Altitude Tropical Stream: The Importance of Season and Flow." *International Review of Hydrobiology* 96, no. 6: 667–685.

Ríos-Touma, B., and A. Ramírez. 2019. "Multiple Stressors in the Neotropical Region: Environmental Impacts in Biodiversity Hotspots." In *Multiple Stressors in River Ecosystems*, 205–220. Elsevier.

Roldán, G. 1988. *Guía para el estudio de los macroinvertebrados acuáticos del Departamento de Antioquia*. Universidad de Antioquia, Fondo FEN.

Sánchez-Argüello, R., A. Cornejo, R. G. Pearson, and L. Boyero. 2010. "Spatial and Temporal Variation of Stream Communities in a Human-Affected Tropical Watershed." *Annales De Limnologie-International Journal of Limnology* 46, no. 3: 149–156.

Serna, D. J., C. E. Tamaris-Turizo, and L. C. Gutiérrez Moreno. 2015. "Distribución espacial y temporal de larvas de Trichoptera (Insecta) en el río Manzanares, Sierra Nevada de Santa Marta (Colombia)." *Revista de Biología Tropical* 63, no. 2: 465–477.

Sinche, F., M. Cabrera, L. Vaca, E. Segura, and P. Carrera. 2023. "Determination of the Ecological Water Quality in the Orienco Stream Using Benthic Macroinvertebrates in the Northern Ecuadorian Amazon." *Integrated Environmental Assessment and Management* 19, no. 3: 615–625.

Sites, R. W., M. R. Willig, and M. J. Linit. 2003. "Macroecology of Aquatic Insects: A Quantitative Analysis of Taxonomic Richness and Composition in the Andes Mountains of Northern Ecuador." *Biotropica* 35, no. 2: 226–239.

Strahler, A. N. 1957. "Quantitative Analysis of Watershed Geomorphology." *Eos, Transactions American Geophysical Union* 38, no. 6: 913–920.

Sumudumali, R. G. I., and J. M. C. K. Jayawardana. 2021. "A Review of Biological Monitoring of Aquatic Ecosystems Approaches: With Special Reference to Macroinvertebrates and Pesticide Pollution." *Environmental Management* 67, no. 2: 263–276.

Tampo, L., I. Kaboré, E. H. Alhassan, A. Ouéda, L. M. Bawa, and G. Djaneye-Boundjou. 2021. "Benthic Macroinvertebrates as Ecological Indicators: Their Sensitivity to the Water Quality and Human Disturbances in a Tropical River." *Frontiers in Water* 3: 662765.

Tomanova, S., E. Goitia, and J. Helešic. 2006. "Trophic Levels and Functional Feeding Groups of Macroinvertebrates in Neotropical Streams." *Hydrobiologia* 556: 251–264.

Torremorell, A., C. Hegoburu, A. L. Brandimarte, et al. 2021. "Current and Future Threats for Ecological Quality Management of South American Freshwater Ecosystems." *Inland Waters* 11, no. 2: 125–140.

Urdanigo, J. P., M. Díaz Ponce, C. Tay-Hing Cajas, et al. 2019. "Diversidad de macroinvertebrados acuáticos en quebradas con diferente cobertura ribereña en el Bosque Protector Murocomba, Ecuador." *Revista de Biología Tropical* 67, no. 4: 861–878.

Vannote, R. L., G. W. Minshall, K. W. Cummins, J. R. Sedell, and C. E. Cushing. 1980. "The River Continuum Concept." *Canadian Journal of Fisheries and Aquatic Sciences* 37, no. 1: 130–137.

Vargas-Tierras, T., S. Suárez-Cedillo, V. Morales-León, et al. 2023. "Ecological River Water Quality Based on Macroinvertebrates Present in the Ecuadorian Amazon." *Sustainability* 15, no. 7: 5790.

Villamarín, C., M. Rieradevall, and N. Prat. 2020. "Macroinvertebrate Diversity Patterns in Tropical Highland Andean Rivers." *Limnetica* 39, no. 2: 677–691.

Villamarín, C., S. Villamarín-Cortez, D. M. Salcido, M. Herrera-Madrid, and B. Ríos-Touma. 2021. "Drivers of Diversity and Altitudinal Distribution of Chironomids (Diptera: Chironomidae) in the Ecuadorian Andes." *Revista de Biología Tropical* 69, no. 1: 113–126.

Wallace, J. B., and J. R. Webster. 1996. "The Role of Macroinvertebrates in Stream Ecosystem Function." *Annual Review of Entomology* 41, no. 1: 115–139.

Walters, K., and L. D. Coen. 2006. "A Comparison of Statistical Approaches to Analyzing Community Convergence Between Natural and Constructed Oyster Reefs." *Journal of Experimental Marine Biology and Ecology* 330: 81–95.

Zhang, Y., and J. S. Richardson. 2011. "Contrasting Effects of Cross-Ecosystem Subsidies and Predation on Benthic Invertebrates in Two Pacific Coastal Streams." *Aquatic Sciences* 73: 53–62.

Zuur, A., E. N. Ieno, and G. M. Smith. 2007. *Analyzing Ecological Data*. Springer.

Zuur, A., E. N. Ieno, N. Walker, A. A. Saveliev, and G. M. Smith. 2009. *Mixed Effects Models and Extensions in Ecology With R*. Springer Science & Business Media.

Supporting Information

Additional supporting information can be found online in the Supporting Information section.